THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\sqrt{74}$	$h^2 = 5^2 + 7^2 = 74$
2	**(d)** 130	Square on hypotenuse $= a^2 + b^2 = 130$
3	**(d)** 16, 63, 65	$2m = 16$, $m^2 - 1 = 63$, $m^2 + 1 = 65$
4	**(d)** right-angled	$14^2 + 48^2 = 2500$ and $50^2 = 2500$
5	**(c)** $3\sqrt{2}$	$d^2 = s^2 + s^2 = 18$
6	**(a)** 210	Other leg $= 21$ cm; area $= \frac{1}{2} \times 20 \times 21$
7	**(b)** the square of the hypotenuse equals the sum of the squares of the other two sides	Statement of the theorem
8	**(a)** $2\sqrt{10}$	$h^2 = 2^2 + 6^2 = 40$
9	**(a)** 290	Square on hypotenuse $= a^2 + b^2 = 290$
10	**(b)** 18, 80, 82	$2m = 18$, $m^2 - 1 = 80$, $m^2 + 1 = 82$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Take $2m = 28$, so $m = 14$ (1); triplet $(28, 195, 197)$ (1).	1+1
12	$16^2 + 63^2 = 4225$ (1); $65^2 = 4225$ so the triangle is right-angled, the right angle is opposite the side 65 cm (1).	1+1
13	$d^2 = 4^2 + 4^2 = 32$ (1); $d = 4\sqrt{2}$ cm (1).	1+1
14	Take $2m = 30$, so $m = 15$ (1); triplet $(30, 224, 226)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x^2 = (4\sqrt{2})^2 = 32$ (1); $x = 4$ cm (1); area $= \frac{1}{2} \times 4 \times 4 = 8$ cm² (1).	1+1+1
16	$d^2 = 29^2 - 21^2 = 841 - 441 = 400$ (2); $d = 20$ m (1).	2+1
17	Diagonals bisect at right angles: half-diagonals 10 cm and 24 cm (1); side$^2 = 100 + 576 = 676$ (1); side $= 26$ cm (1).	1+1+1
	OR: $2x^2 = (6\sqrt{2})^2 = 72$ (1); $x = 6$ cm (1); area $= \frac{1}{2} \times 6 \times 6 = 18$ cm² (1).	1+1+1
18	$d^2 = 41^2 - 40^2 = 1681 - 1600 = 81$ (2); $d = 9$ m (1).	2+1
	OR: Diagonals bisect at right angles: half-diagonals 12 cm and 35 cm (1); side$^2 = 144 + 1225 = 1369$ (1); side $= 37$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-diagonals 20 cm, 21 cm (1); side$^2 = 400 + 441 = 841$, side $= 29$ cm (2); perimeter $= 4 \times 29 = 116$ cm (1); area $= \frac{1}{2} \times 40 \times 42 = 840$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 20 - 12 = 8$ m (1); distance$^2 = 8^2 + 15^2 = 289$ (2); distance $= 17$ m (2).	1+2+2
20	Half-diagonals 9 cm, 40 cm (1); side$^2 = 81 + 1600 = 1681$, side $= 41$ cm (2); perimeter $= 4 \times 41 = 164$ cm (1); area $= \frac{1}{2} \times 18 \times 80 = 720$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 30 - 6 = 24$ m (1); distance$^2 = 24^2 + 10^2 = 676$ (2); distance $= 26$ m (2).	1+2+2

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** not right-angled	$11^2 + 25^2 = 746$ and $30^2 = 900$
2	**(d)** $5\sqrt{2}$	$d^2 = s^2 + s^2 = 50$
3	**(d)** 180	Other leg $= 40$ cm; area $= \frac{1}{2} \times 9 \times 40$
4	**(c)** right-angled	Converse of the theorem
5	**(c)** $\sqrt{34}$	$h^2 = 3^2 + 5^2 = 34$
6	**(c)** 221	Square on hypotenuse $= a^2 + b^2 = 221$
7	**(a)** 20, 99, 101	$2m = 20$, $m^2 - 1 = 99$, $m^2 + 1 = 101$
8	**(d)** right-angled	$24^2 + 45^2 = 2601$ and $51^2 = 2601$
9	**(d)** $6\sqrt{2}$	$d^2 = s^2 + s^2 = 72$
10	**(d)** 120	Other leg $= 24$ cm; area $= \frac{1}{2} \times 10 \times 24$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$18^2 + 24^2 = 900 \neq 32^2 = 1024$ (1) so it is not a right triangle (1).	1+1
12	$d^2 = 8^2 + 8^2 = 128$ (1); $d = 8\sqrt{2}$ cm (1).	1+1
13	Take $2m = 32$, so $m = 16$ (1); triplet $(32, 255, 257)$ (1).	1+1
14	$45^2 + 108^2 = 13689$ (1); $117^2 = 13689$ so the triangle is right-angled, the right angle is opposite the side 117 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x^2 = (7\sqrt{2})^2 = 98$ (1); $x = 7$ cm (1); area $= \frac{1}{2} \times 7 \times 7 = 24.5$ cm² (1).	1+1+1
16	$d^2 = 26^2 - 24^2 = 676 - 576 = 100$ (2); $d = 10$ m (1).	2+1
17	Diagonals bisect at right angles: half-diagonals 11 cm and 60 cm (1); side$^2 = 121 + 3600 = 3721$ (1); side $= 61$ cm (1).	1+1+1
	OR: $2x^2 = (9\sqrt{2})^2 = 162$ (1); $x = 9$ cm (1); area $= \frac{1}{2} \times 9 \times 9 = 40.5$ cm² (1).	1+1+1
18	$d^2 = 37^2 - 35^2 = 1369 - 1225 = 144$ (2); $d = 12$ m (1).	2+1
	OR: Diagonals bisect at right angles: half-diagonals 16 cm and 30 cm (1); side$^2 = 256 + 900 = 1156$ (1); side $= 34$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-diagonals 10 cm, 24 cm (1); side$^2 = 100 + 576 = 676$, side $= 26$ cm (2); perimeter $= 4 \times 26 = 104$ cm (1); area $= \frac{1}{2} \times 20 \times 48 = 480$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 17 - 5 = 12$ m (1); distance$^2 = 12^2 + 16^2 = 400$ (2); distance $= 20$ m (2).	1+2+2
20	Half-diagonals 12 cm, 35 cm (1); side$^2 = 144 + 1225 = 1369$, side $= 37$ cm (2); perimeter $= 4 \times 37 = 148$ cm (1); area $= \frac{1}{2} \times 24 \times 70 = 840$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 45 - 5 = 40$ m (1); distance$^2 = 40^2 + 42^2 = 3364$ (2); distance $= 58$ m (2).	1+2+2

Mistakes to avoid:

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- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** congruent right triangles	Rectangle has four right angles
2	**(d)** $2\sqrt{13}$	$h^2 = 4^2 + 6^2 = 52$
3	**(a)** 353	Square on hypotenuse $= a^2 + b^2 = 353$
4	**(c)** 22, 120, 122	$2m = 22$, $m^2 - 1 = 120$, $m^2 + 1 = 122$
5	**(c)** not right-angled	$13^2 + 16^2 = 425$ and $22^2 = 484$
6	**(a)** $7\sqrt{2}$	$d^2 = s^2 + s^2 = 98$
7	**(b)** 210	Other leg $= 35$ cm; area $= \frac{1}{2} \times 12 \times 35$
8	**(b)** $5\sqrt{2}$	$h^2 = 1^2 + 7^2 = 50$
9	**(b)** 505	Square on hypotenuse $= a^2 + b^2 = 505$
10	**(b)** 24, 143, 145	$2m = 24$, $m^2 - 1 = 143$, $m^2 + 1 = 145$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d^2 = 10^2 + 10^2 = 200$ (1); $d = 10\sqrt{2}$ cm (1).	1+1
12	Take $2m = 36$, so $m = 18$ (1); triplet $(36, 323, 325)$ (1).	1+1
13	$19^2 + 21^2 = 802 \neq 29^2 = 841$ (1) so it is not a right triangle (1).	1+1
14	$d^2 = 12^2 + 12^2 = 288$ (1); $d = 12\sqrt{2}$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x^2 = (10\sqrt{2})^2 = 200$ (1); $x = 10$ cm (1); area $= \frac{1}{2} \times 10 \times 10 = 50$ cm² (1).	1+1+1
16	$d^2 = 61^2 - 60^2 = 3721 - 3600 = 121$ (2); $d = 11$ m (1).	2+1
17	Diagonals bisect at right angles: half-diagonals 28 cm and 45 cm (1); side$^2 = 784 + 2025 = 2809$ (1); side $= 53$ cm (1).	1+1+1
	OR: $2x^2 = (12\sqrt{2})^2 = 288$ (1); $x = 12$ cm (1); area $= \frac{1}{2} \times 12 \times 12 = 72$ cm² (1).	1+1+1
18	$d^2 = 34^2 - 30^2 = 1156 - 900 = 256$ (2); $d = 16$ m (1).	2+1
	OR: Diagonals bisect at right angles: half-diagonals 33 cm and 56 cm (1); side$^2 = 1089 + 3136 = 4225$ (1); side $= 65$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Half-diagonals 11 cm, 60 cm (1); side$^2 = 121 + 3600 = 3721$, side $= 61$ cm (2); perimeter $= 4 \times 61 = 244$ cm (1); area $= \frac{1}{2} \times 22 \times 120 = 1320$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 35 - 11 = 24$ m (1); distance$^2 = 24^2 + 32^2 = 1600$ (2); distance $= 40$ m (2).	1+2+2
20	Half-diagonals 16 cm, 30 cm (1); side$^2 = 256 + 900 = 1156$, side $= 34$ cm (2); perimeter $= 4 \times 34 = 136$ cm (1); area $= \frac{1}{2} \times 32 \times 60 = 960$ cm² (1).	1+2+1+1
	OR: Difference in heights $= 28 - 4 = 24$ m (1); distance$^2 = 24^2 + 7^2 = 625$ (2); distance $= 25$ m (2).	1+2+2

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